Industrialization

Week 5

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Today

Three papers that fit together as an argument.

1. **The claim.** Allen. Britain had the world's dearest labor and its cheapest energy, and that price structure is what called inventions into being.

2. **The test.** Hanlon. Allen's mechanism requires that invention chases relative prices. A civil war on another continent lets us watch it happen.

3. **The puzzle.** Clark. The machines then diffused everywhere, but productivity did not follow. Why not?

Where does inventive activity come from, the supply side or the demand side? Hold that question all evening.

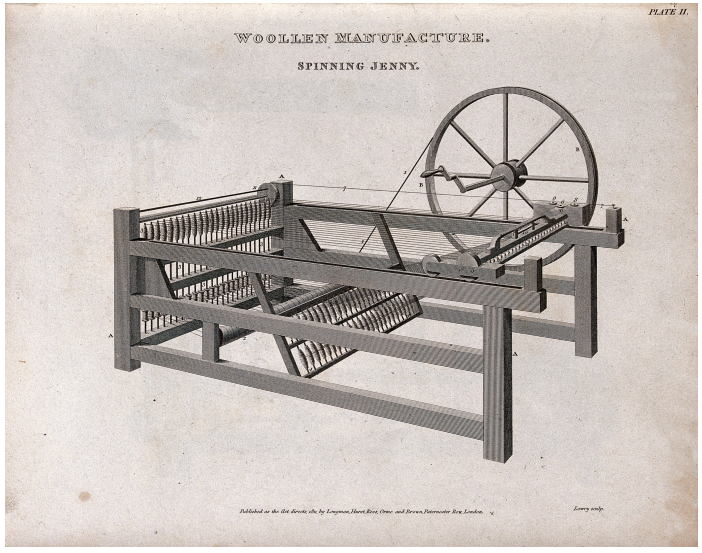
Why Britain



Philip James de Loutherbourg, *Coalbrookdale by Night*, 1801. The Bedlam furnaces in the Severn gorge, a few miles from Darby's works. Science Museum, London, via Wikimedia Commons, public domain.

Invention was everywhere

- The spinning jenny (Hargreaves, about 1764). The steam engine (Newcomen 1712, Watt 1765–75). Coke smelting (Darby 1709).
- The tempting question is why the British were clever. It is the wrong question, because invention was everywhere. Jacquard's loom, the Montgolfiers' balloon, Lavoisier's oxygen — all French, all in the same decades. Earlier centuries gave us spectacles, water wheels, the lateen sail, the stirrup, cannon.
- The right question is why these particular inventions were developed, refined and paid for in Britain, and used there first.



Hargreaves' spinning jenny. Engraving by W. Lowry, 1811. Wellcome Library, London, CC BY 4.0.

> Allen takes Edison's arithmetic seriously. Invention is one percent inspiration and ninety-nine percent perspiration — and perspiration is research and development, which somebody has to fund and expect to profit from.

Allen (2011), "Why the industrial revolution was British."

Three words we will use all evening

Invention and innovation, after Schumpeter

Invention is the act of working out that something can be done. **Innovation** is the first commercially successful application of it. They can be decades apart, and the second is the one that shows up in the growth figures.

Macro- and micro-invention, after Mokyr

A **macro-invention** is a radical new idea without clear precedent, arriving more or less from nowhere. A **micro-invention** is a small incremental step that improves or adapts a technique already in use — cutting cost, improving durability, saving fuel or materials.

> *"The distinction between micro- and macro-inventions matters because they appeared to be governed by different laws."*

Mokyr, *The Lever of Riches* (1990). Micro-inventions come from deliberate search and respond to economic incentives. Macro-inventions look more like genius and luck. And they are **complements** — each makes the other more productive.

What Allen's theory covers

Allen's entire argument is a theory of micro-invention. Prices tell people which improvements are worth the effort of searching for. Hanlon, later tonight, will measure exactly that response.

Neither of them explains where the macro-invention comes from. For that we need the other half of Mokyr's story, which is knowledge.

- So let us take one machine and run it through all three distinctions.

- The steam engine is an instructive choice, because we can watch the knowledge arrive, the macro-invention happen twice, the micro-inventions accumulate for a century, and the innovation lag in between.

The knowledge came from the Continent

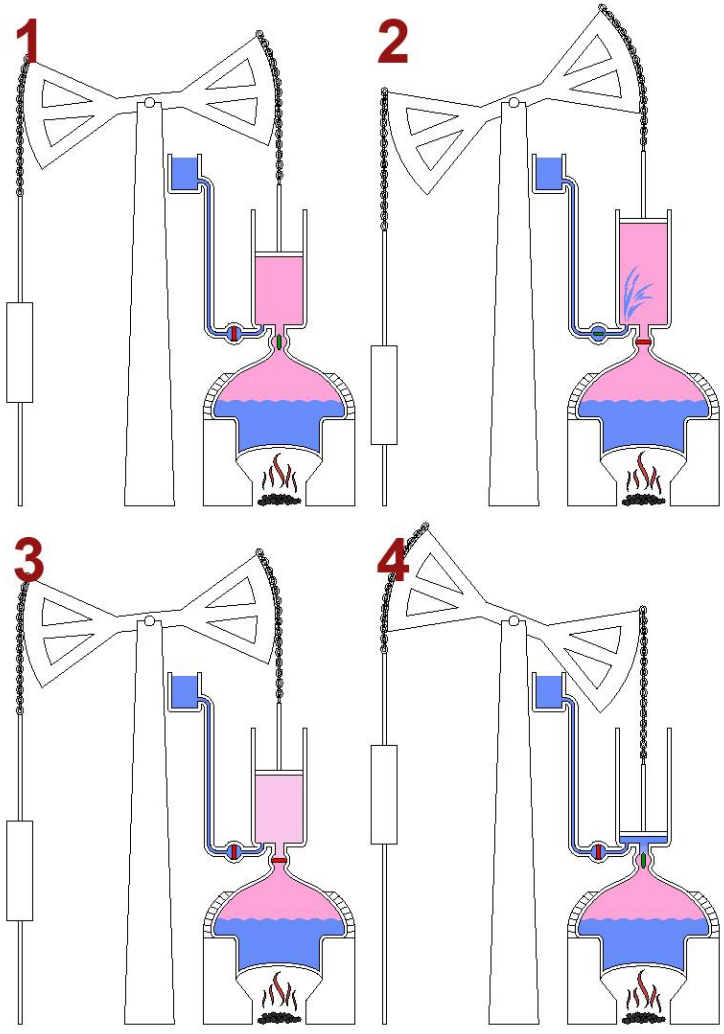


Guericke's hemispheres and the horses. Gaspar Schott's plate in *Experimenta Nova Magdeburgica*, 1672. Wikimedia Commons, public domain.

- You cannot build an engine driven by atmospheric pressure until somebody establishes two things that nobody believed in 1600. That a vacuum can exist and that air has weight.
- Torricelli's barometer, 1643. Pascal having one carried up the Puy de Dôme to show the column falls with altitude, 1648. And Otto von Guericke, mayor of Magdeburg, who built the first working air pump around 1650.
- In the 1650s Guericke evacuated two copper hemispheres, and teams of horses could not pull them apart. Nothing held them together except the pressure of the air.

This is Mokyr's other engine — the stock of **useful knowledge**. Guericke was not trying to pump water out of a coal mine. He was settling an argument with Aristotle. Sixty years later somebody used the answer.

Newcomen's engine, 1712



1. Steam fills the cylinder.
2. Cold water is sprayed in.
3. The steam condenses, leaving a vacuum.
4. Air pressure drives the piston down.

- Steam does not push the piston. It is condensed to make a vacuum, and the weight of the atmosphere does the work — which is why it is called an **atmospheric** engine, and why Guericke had to come first.

Animation by Emoscopes, Wikimedia Commons, CC BY 2.5.

Both animations play at momentousbritain.co.uk.

Why it was a terrible engine, and why nobody minded

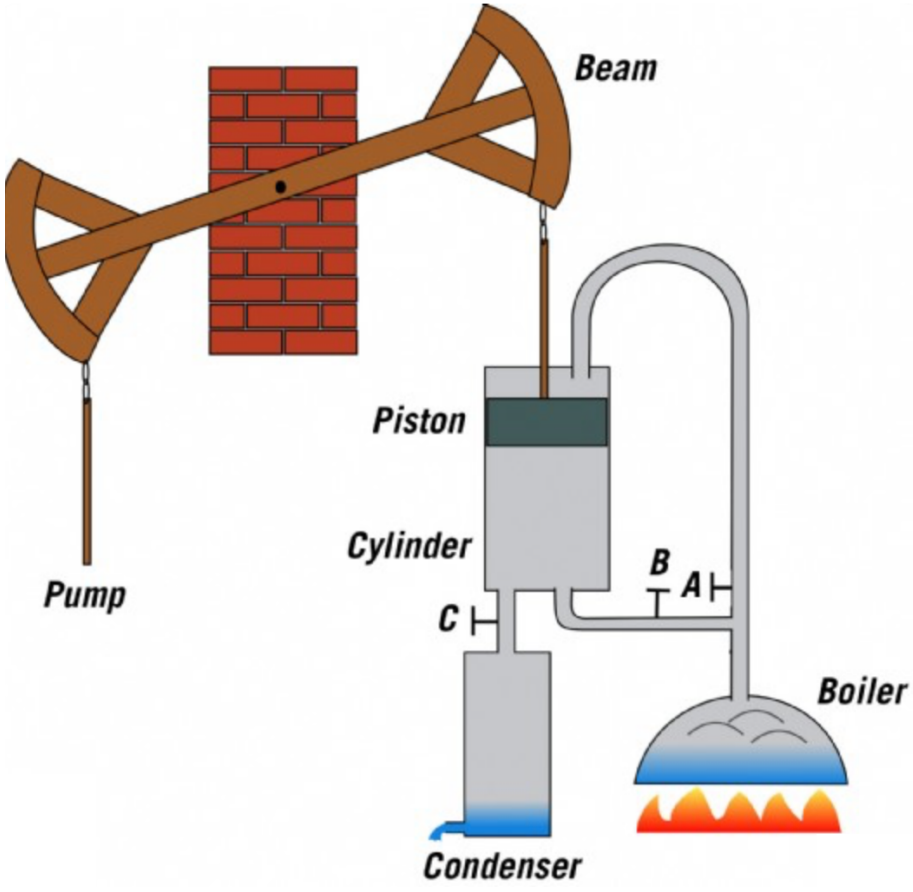
Look again at stages one and two. The same cylinder is filled with steam, then sprayed with cold water. Every single stroke.

- So every stroke you reheat several hundred pounds of iron that you have just deliberately chilled. Almost all the fuel goes into warming the cylinder back up rather than into lifting the beam.

- Which was tolerable in exactly one place — at the mouth of a coal pit, where the fuel was the unsaleable small coal lying on the ground and the engine was pumping water out of the mine that produced it.

A machine whose economics work in one location only. Newcomen engines spread across British coalfields for fifty years and hardly anywhere else, which is Allen's argument in miniature — and it is also the problem Watt solved.

Watt's idea, 1765



- Steam from the boiler enters through valve A. The stroke that matters comes next, when valve C opens and the steam rushes out of the cylinder into a second vessel kept permanently cold.
- Newcomen condensed inside the cylinder and paid for it every stroke. Condensing somewhere else lets the cylinder stay hot forever.

A century of fuel saving followed. Allen puts it at 44 pounds of coal per horsepower-hour in 1727 and 2 pounds in 1860, through Smeaton, Watt, Trevithick and the Cornish mine owners.

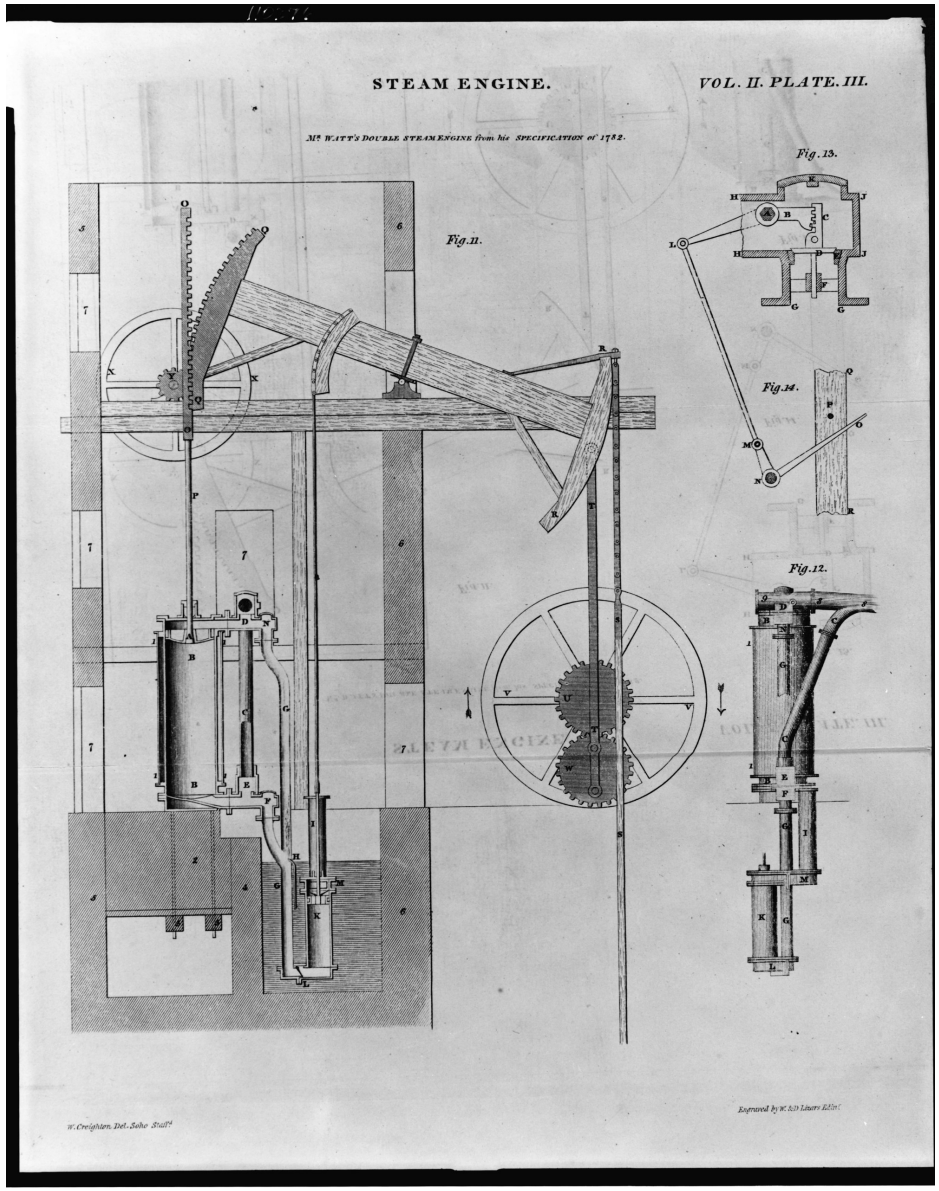
Ten years stood between the invention and the innovation

1765	Watt works out the separate condenser — the invention
1767	partnership with Roebuck
1769	patent granted, January
1774	Boulton acquires Roebuck's share
1775	Boulton and Watt, patent extended by Parliament to 1800
1782	double-acting engine

- **Ten years** from the idea to a business that could actually build and sell the thing, and it took Birmingham capital and Birmingham metalworking to get there.

- The macro-invention was 1765. The innovation was 1775. And what happened between 1775 and 1800 — and for a century after — was micro-invention, thousands of small improvements in valves, cylinders, boring tolerances, governors and gearing.

Watt's double-acting engine, from the 1782 specification



Watt's double-acting engine as specified in 1782. Engraving in John Robison, *A System of Mechanical Philosophy*, Edinburgh, 1822. The cylinder stands at the left with the condenser and air pump in the cistern beneath it, and the beam drives the flywheel through the sun-and-planet gear at the right. Library of Congress, public domain.

What the steam engine teaches

1. **Knowledge first.** No vacuum, no atmospheric engine. Guericke was not solving an economic problem and could not have been.
2. **Macro-inventions are lumpy and rare.** Two in a century — the atmospheric engine and the separate condenser.
3. **Micro-inventions do the work.** They are where most of the efficiency came from, and they are the ones that answer to prices.
4. **Invention is not innovation.** Ten years and a partnership stood between Watt's idea and anything that mattered economically.

Keep the third one in view. When Allen says British prices called forth British inventions, and when Hanlon counts patents after the blockade, they are both talking about the third row of this list.

What made the arithmetic work

- A machine that replaces labor with capital and fuel is worth developing only where labor is expensive relative to capital and fuel.
- That is a claim about a price structure and prices are measurable.
- So Allen's argument stands or falls on five facts.

Notice the shape of the argument. It is **demand-side**. The supply of ideas is taken as roughly common across Europe. What differs is the incentive to turn an idea into a working, profitable machine.

Fact one — British wages were the highest in the world

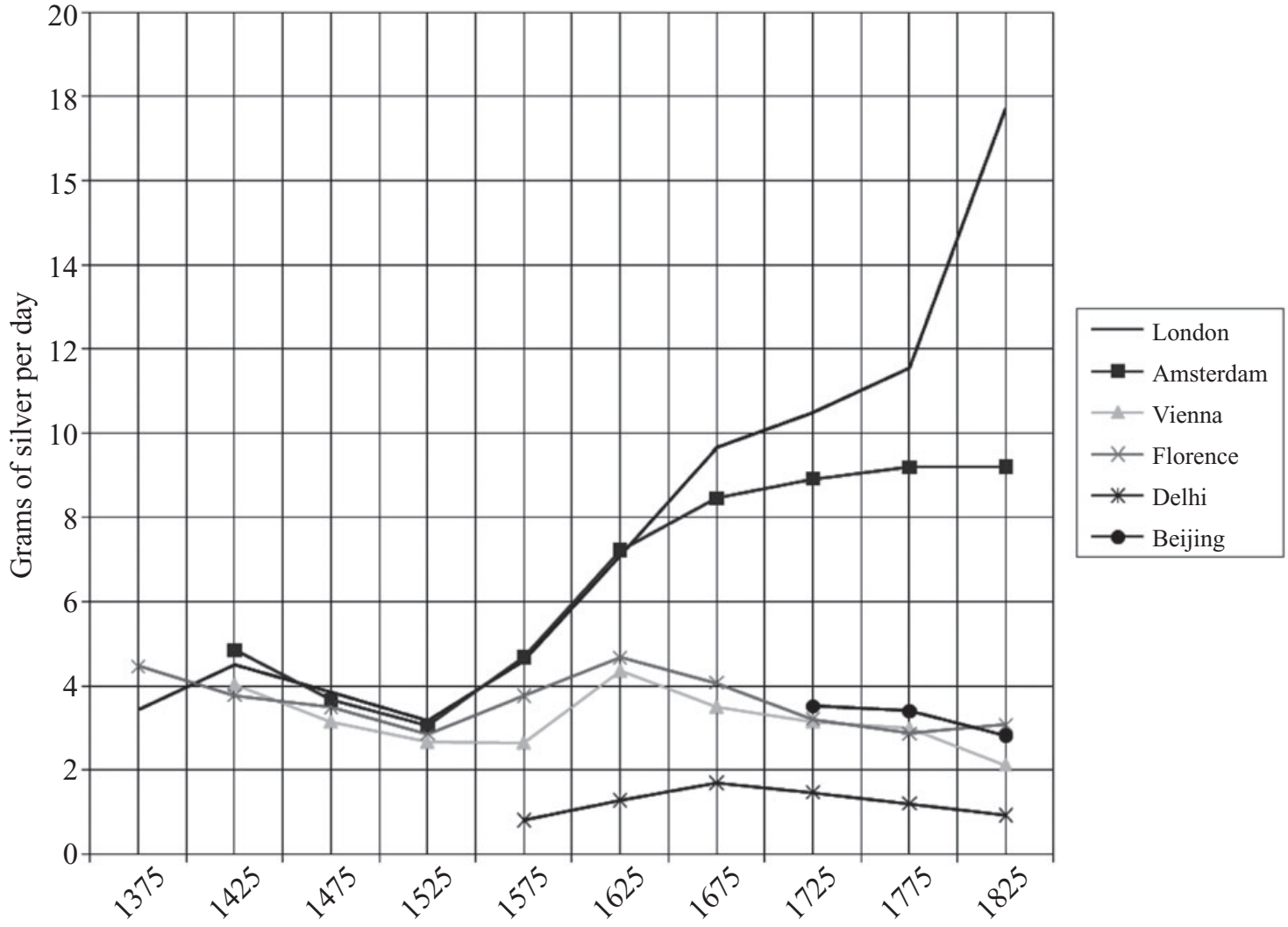


Figure 1 *Labourers' wages around the world*

Allen (2011), Figure 1. Nominal wages of building laborers in grams of silver per day. London and Amsterdam separate from the field after 1600. Delhi and Beijing sit near the bottom throughout.

Fact two — and they were high in real terms too

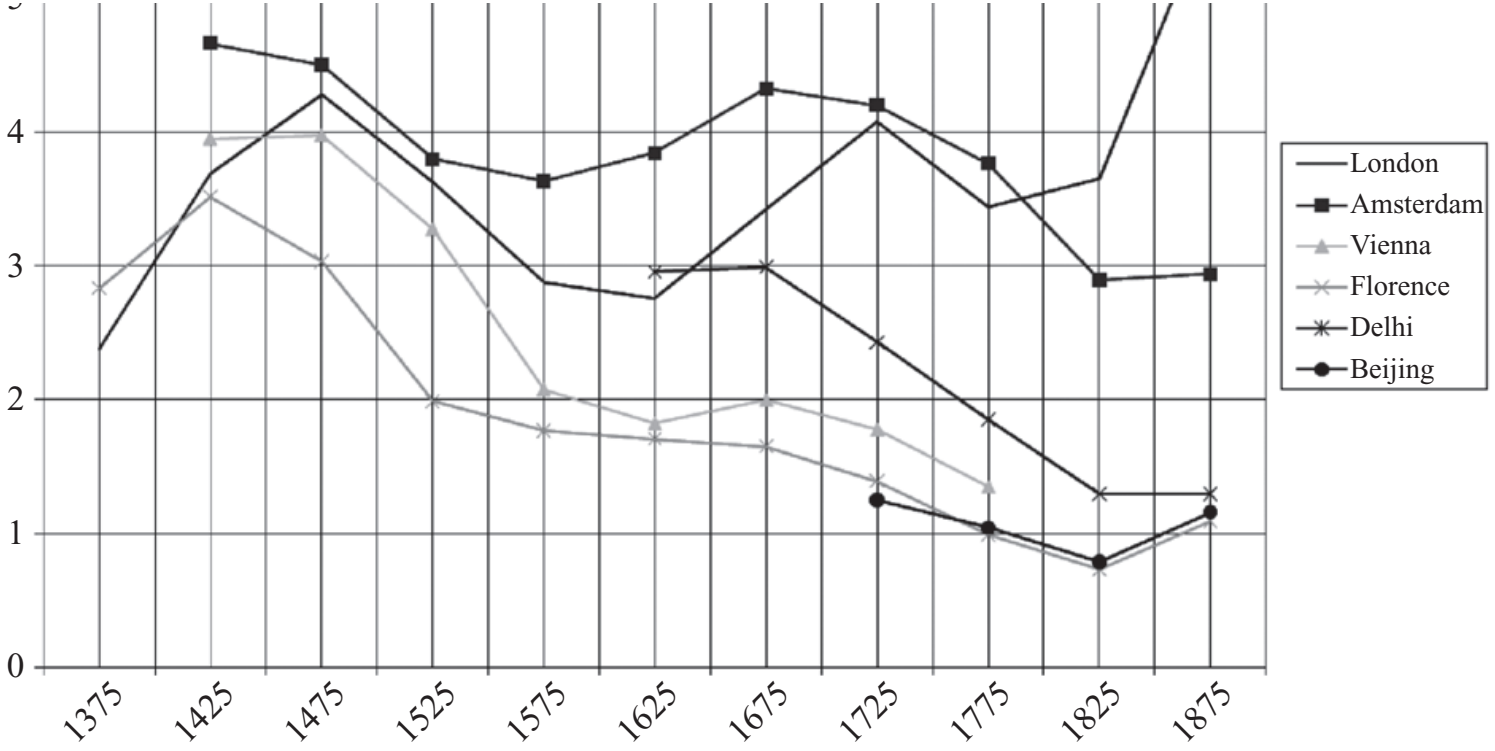


Figure 2. *Subsistence ratio for labourers: income relative to cost of subsistence baske*

Source: See text.

Allen (2011), Figure 2. Welfare ratios — annual earnings divided by the cost of a subsistence basket. London and Amsterdam hold three to five times subsistence while Delhi and Beijing fall toward one. The wage advantage survives deflating by the cost of living.

Fact three — labor was dear relative to capital

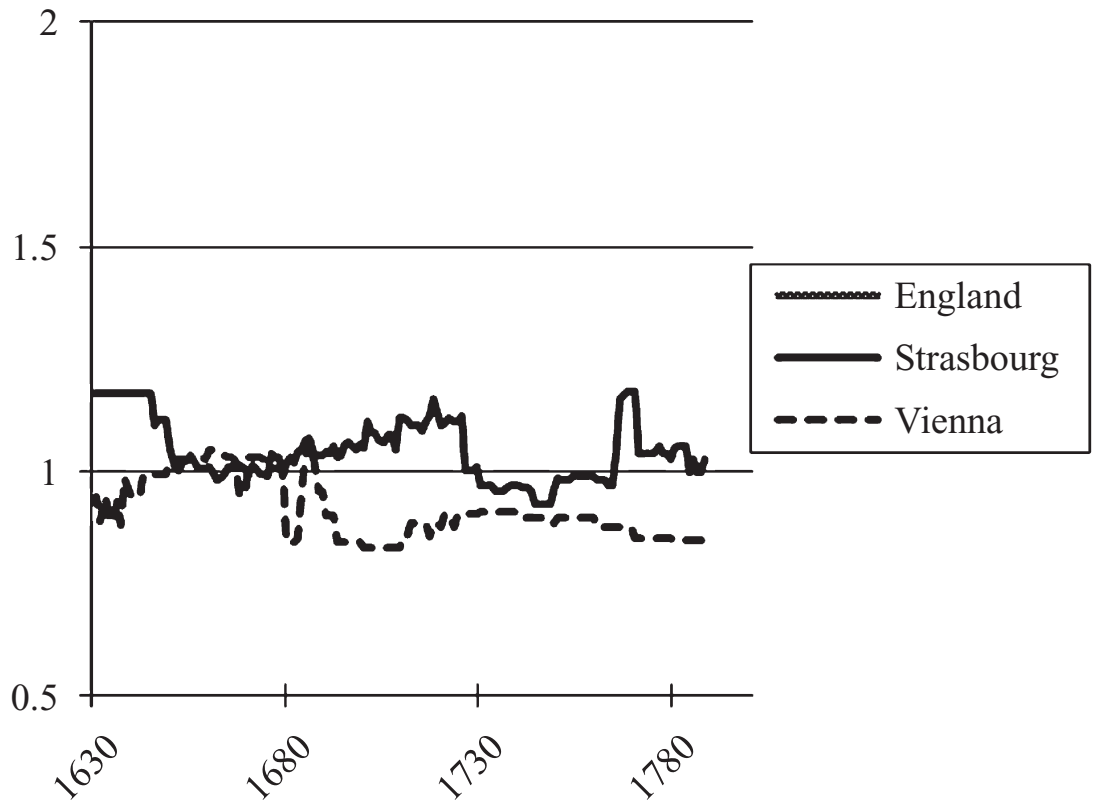


Figure 3. *Wage rate relative to price of capital*
Source: See text.

Allen (2011), Figure 3. The wage relative to the rental price of capital, 1630–1790. The England series is missing from the figure as printed in the journal. Allen's text describes labor growing steadily dearer relative to capital in England from 1650 while the ratio declined in Strasbourg and Vienna. Vienna ends near 0.85. This is the ratio the isoquant diagram will turn on.

Fact four — British energy was the cheapest in the world

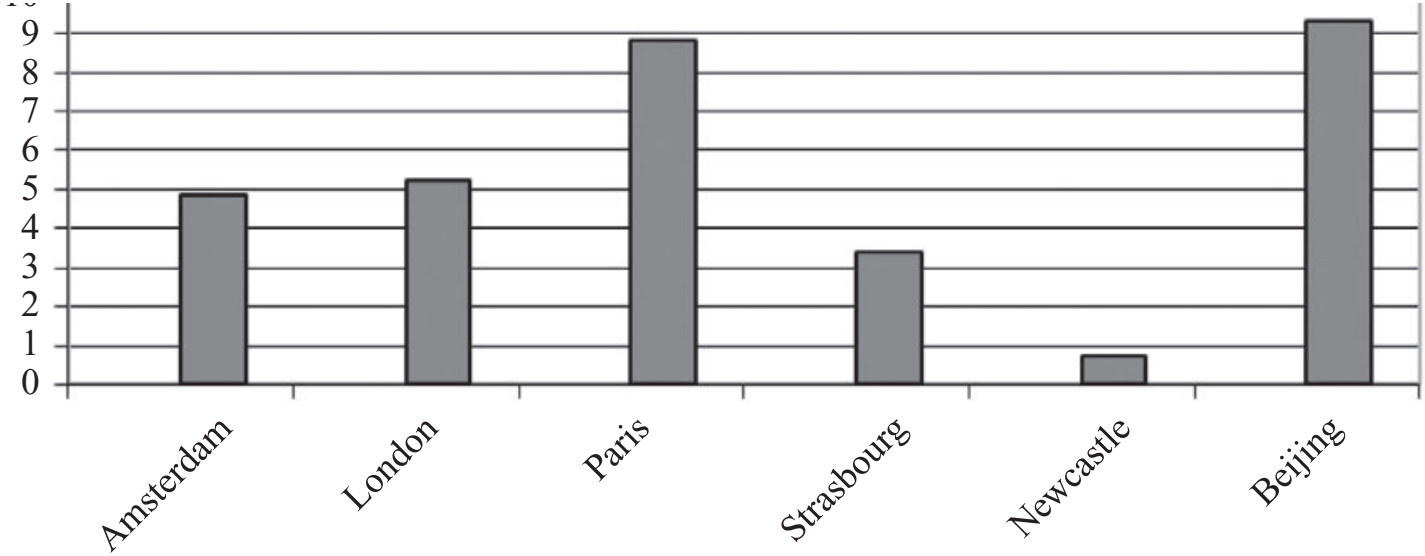
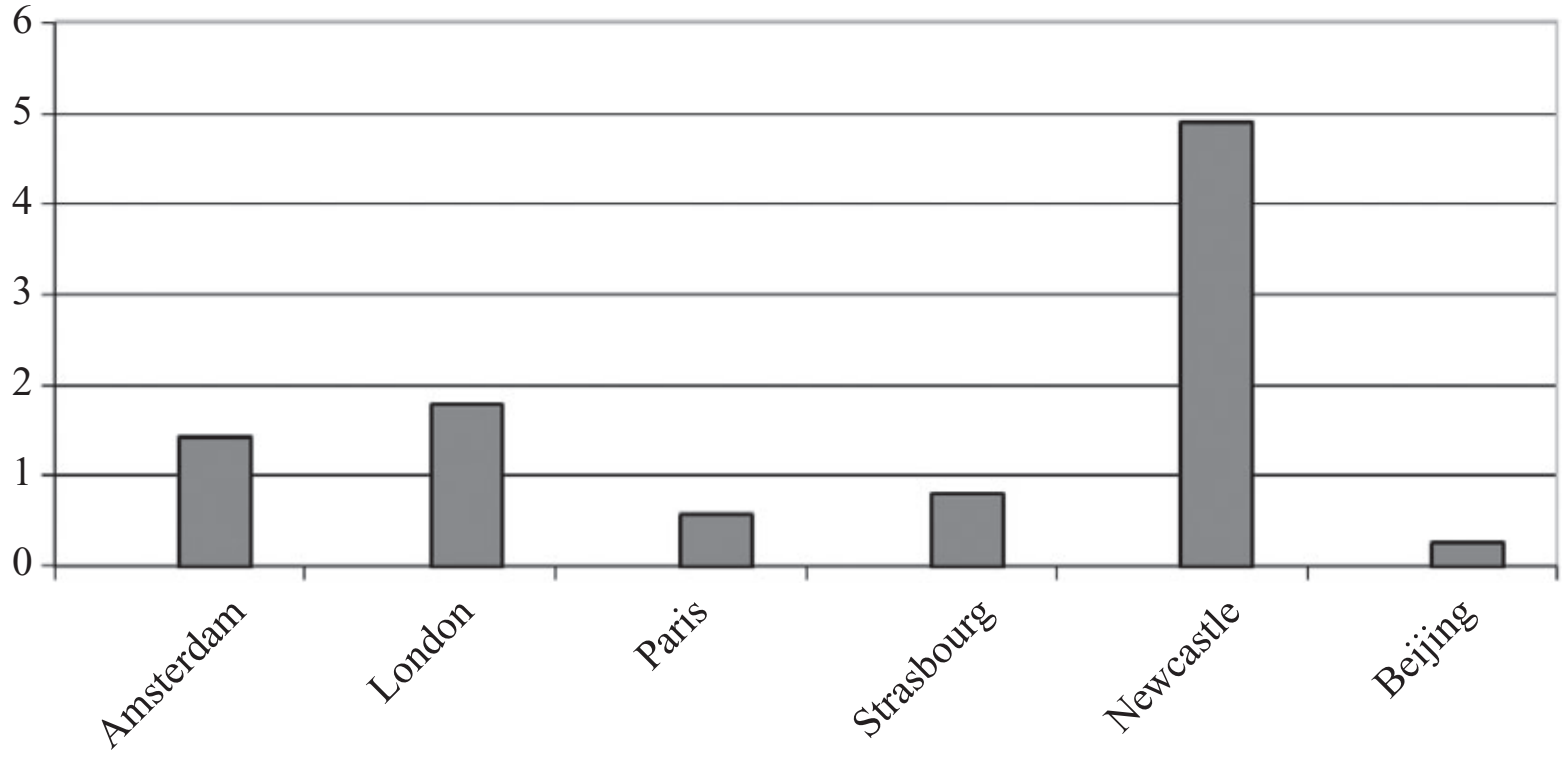


Figure 5. *Price of energy, early 1700s*

Source: See text.

Allen (2011), Figure 5. Price of energy, early 1700s. Newcastle sits on its own coal and is an order of magnitude below Paris and Beijing. London burned Newcastle coal shipped down the coast, and transport was most of its price, which is why London sits mid-table rather than beside Newcastle.

Fact five — the wage relative to energy was highest in Britain



Allen (2011), Figure 4. The wage rate relative to the price of energy. Newcastle is roughly three times London and nearly twenty times Beijing. This is the number a would-be inventor of a steam engine was facing.

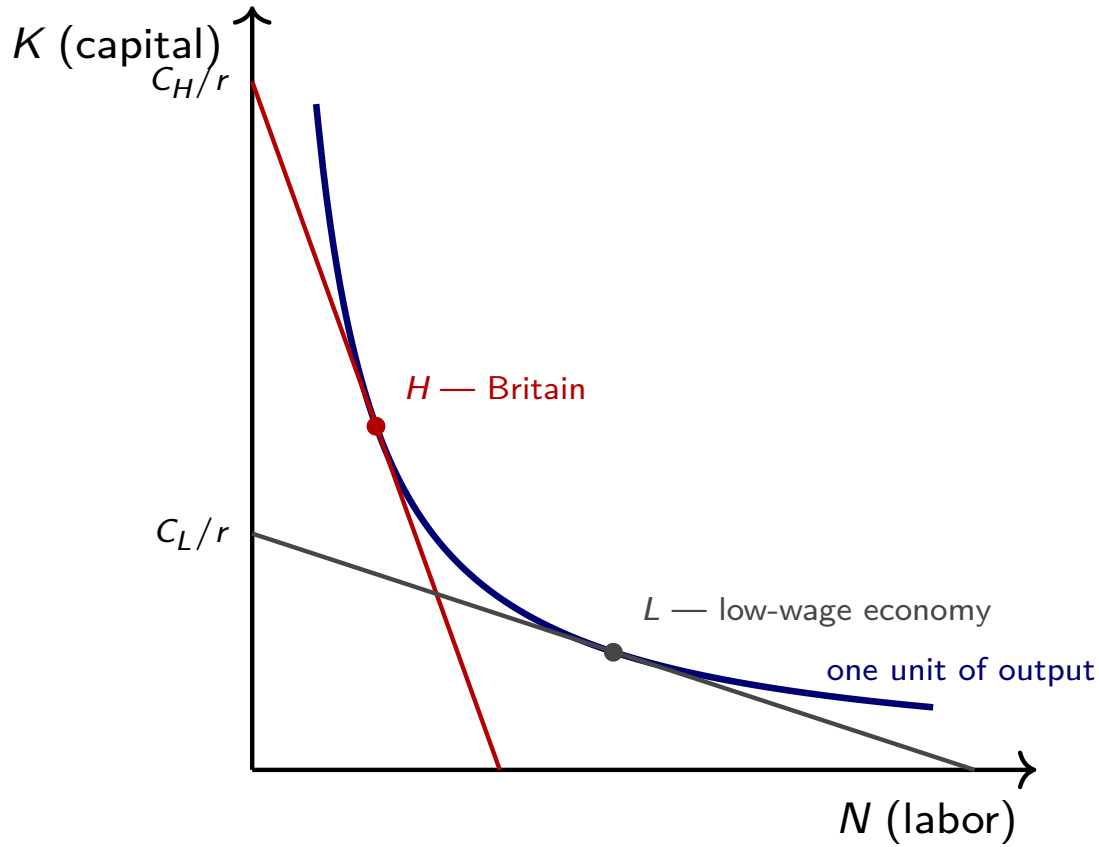
Where that price structure came from

Geology was only half of it. Allen traces the wage half to commerce and empire.

- Light woolen cloth exports from the sixteenth century, then mercantilist and imperial policy protecting and extending that trade.
- Success in international trade pulled labor into cities and into high-productivity work, and bid wages up.
- High wages then fed back into the supply of invention as well as the demand for it. Well-paid people could afford literacy, numeracy and apprenticeships, so Britain had an unusually large stock of skilled artisans capable of building what was designed.

So the state-and-commerce story from Weeks 3 and 4 is upstream of the technology story. Mercantilism and empire set the prices that made the machines worth inventing.

The logic in one diagram



- Each straight line is a set of equal-cost input combinations, $K = C/r - (w/r)N$. Its slope is the wage relative to the rental price of capital, so a high-wage economy faces a steep line.
- Britain's steep line is tangent where the isoquant is **capital intensive**. A technique using more machinery and less labor is the cheap way to produce there, and only there.

Redrawn from Allen (2011), Figure 6.

How Allen actually argues it

There is no regression in this paper. Not one coefficient, not one standard error, and the argument is complete without one.

His argument has three moving parts instead.

1. **Measurement.** Assemble comparable wage and price series for a dozen cities over four centuries, in grams of silver and in baskets of goods. Most of the labor in the paper is here.

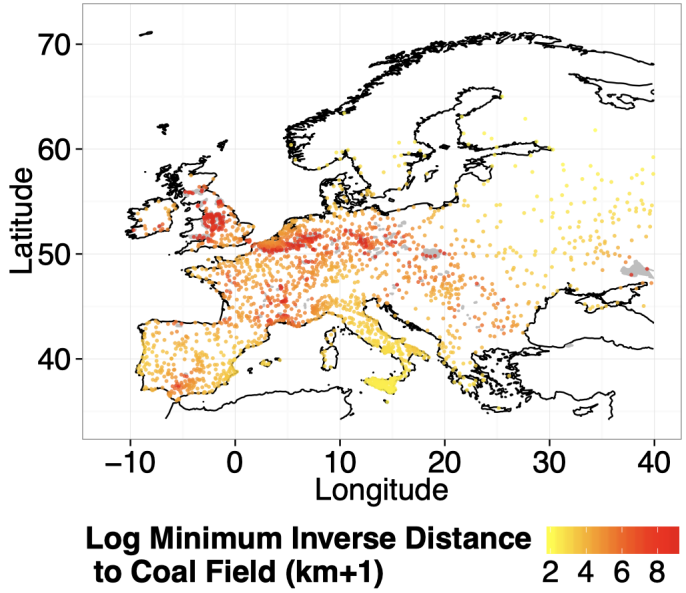
2. **A model.** The isoquant diagram. Given w/r and w/p_{energy}, which techniques are worth developing?

3. **A profitability calculation.** Take a specific machine, price it at British factor prices and at French or Indian ones, and ask where the rate of return clears the hurdle.

The identification, such as it is, comes from cross-sectional variation in prices plus a theory of what firms do with prices. That is a real argument, and it is also exactly the kind of argument Week 3 taught you to be careful with.

The two objections

- **Why not the Dutch?** They had wages nearly as high. But their fuel was peat, and Newcastle coal could be shipped into the Low Countries cheaply enough to remove any urgency about opening the Ruhr coalfields. The incentive needed dear labor and cheap coal together.
- **Why not sooner?** Because the knowledge was not there, which is the Guericke point from earlier. Cheap coal and dear labor cannot call forth an atmospheric engine in 1600, since in 1600 nobody can tell you that the atmosphere presses on things.



European cities colored by log inverse distance to the nearest coalfield. Fernihough and O'Rourke, "Coal and the European Industrial Revolution," *Economic Journal*, 2021.

Belgium had coal too, and used it. But Belgian output in 1800 was about **13 percent** of Britain's, so the return to inventing coal-using technology there was correspondingly smaller. The argument is always about magnitudes.

Micro-improvement carries the technique past the tipping point

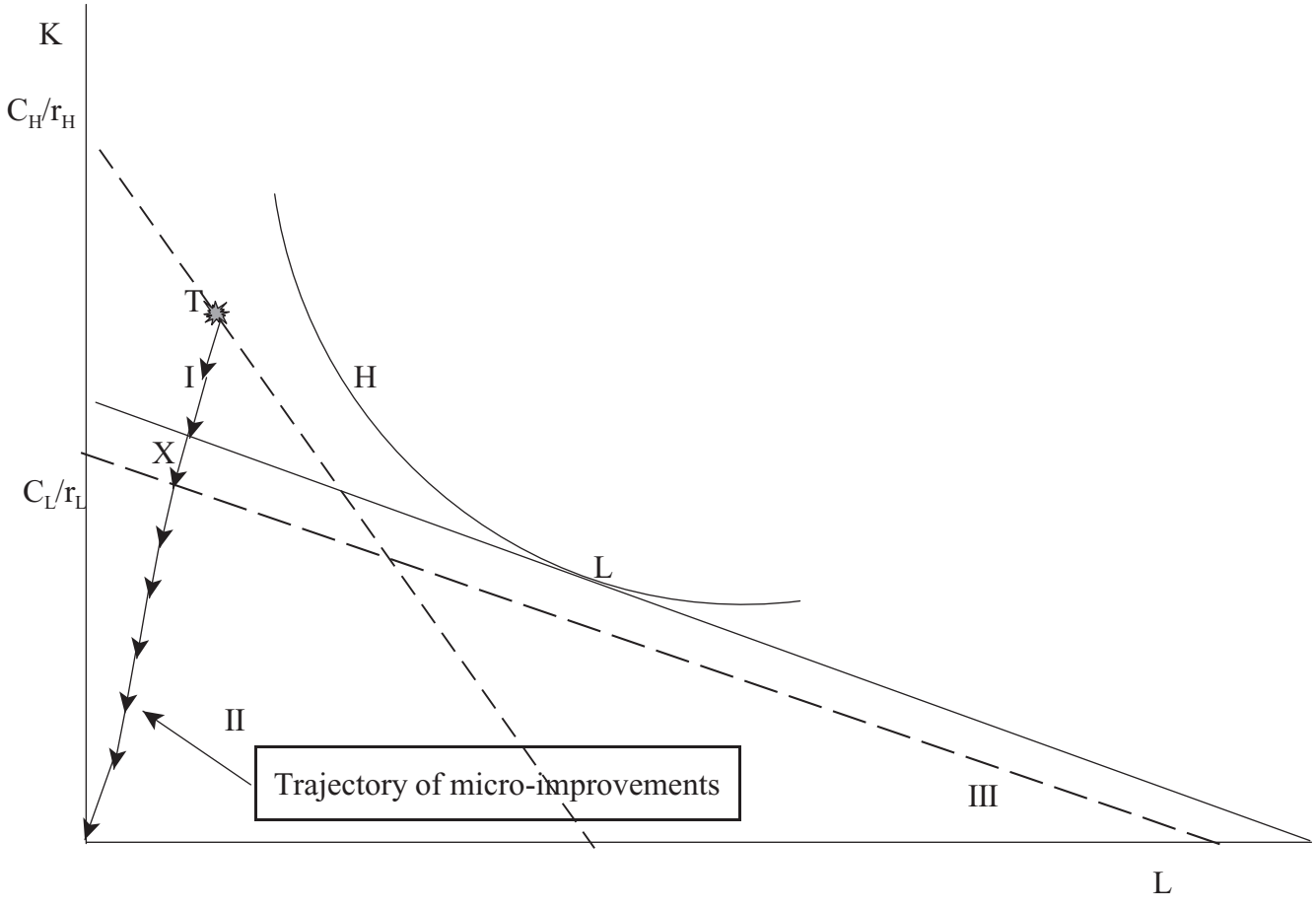


Figure 7. *The trajectory of micro-improvements*
Source: See text.

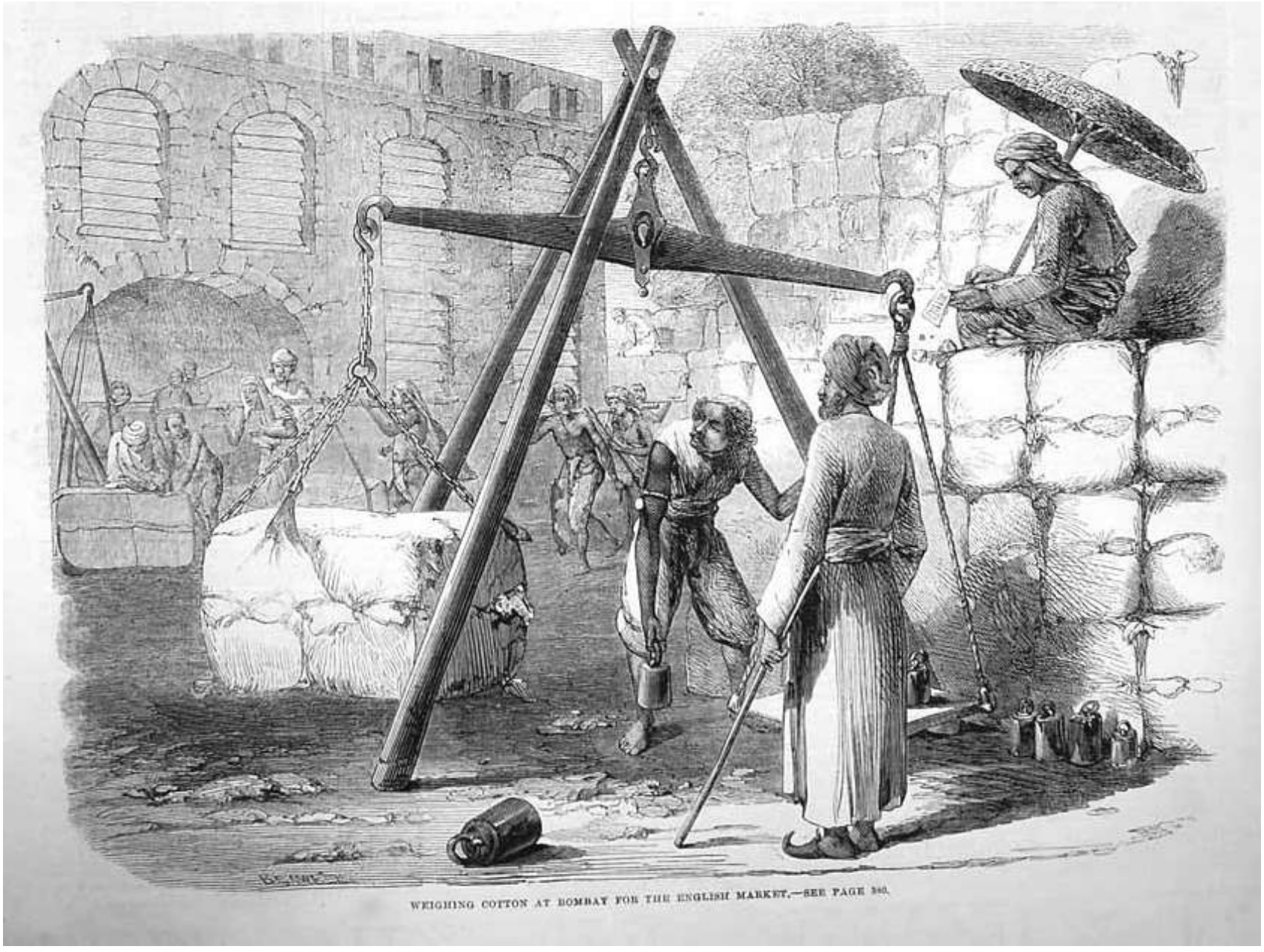
Allen (2011), Figure 7. The new technique T pays only on Britain's steep isocost line. Micro-improvement drags it toward the origin until, at X, it pays at foreign prices too.

The tipping point

The technologies were invented where they were profitable to invent, and at first they were profitable nowhere else. Then decades of micro-improvement made them cheap enough to pay at foreign prices too — and when that tipping point arrived, other countries adopted them in their most advanced form.

- That is a satisfying story, and it makes two commitments we can check.
- First, that invention really does chase relative prices. Allen cannot test that in the eighteenth century, because there is no counterfactual Britain with cheap labor. Hanlon can.
- Second, that once the machines were cheap enough, adoption would spread the productivity with them. Clark says look again.

Does invention chase prices?



"Weighing cotton at Bombay for the English market." *Illustrated London News*, 1862. Wikimedia Commons, public domain.

Allen's claim as an identification problem

- The claim is a comparative static. Change relative input prices, and the direction of technical change changes.

- To test it you would want a sudden, large, exogenous shift in the relative supply of an input, in an industry where you can measure innovation before and after, with a plausible control group of technologies that the shock did not touch.

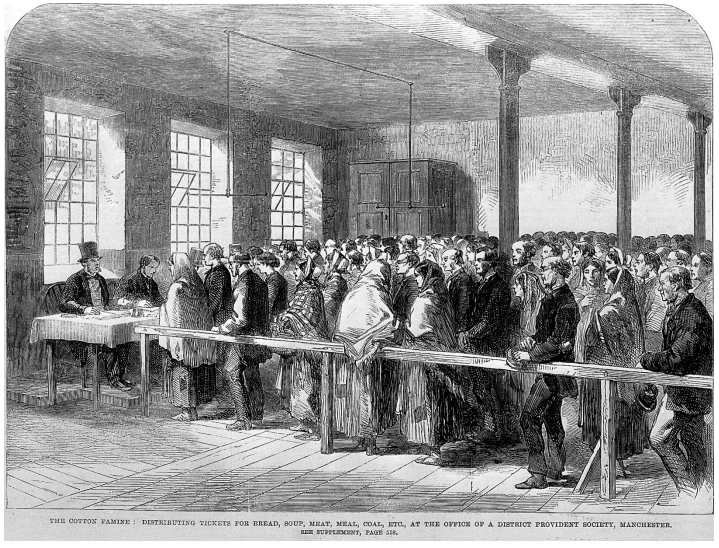
- You cannot run that experiment. But in April 1861 history ran it for us.

> This is the Week 2 checklist again. What is the shock, is it plausibly unrelated to the outcome except through the channel claimed, and what is the comparison?

Hanlon (2015), "Necessity Is the Mother of Invention."

The experiment

- The Union blockade of Southern ports cut off the supply of American cotton to Britain, which was running the largest manufacturing industry on earth almost entirely on it.
- British cotton output fell by as much as **half**. Hundreds of thousands of Lancashire operatives were thrown out of work or onto short time. It is remembered as the **cotton famine**.
- Producers switched to what they could get, which was Indian cotton, mostly **Surat**.



Relief tickets for bread, soup, meat, meal and coal at a District Provident Society office, Manchester. *Illustrated London News*, 22 November 1862. Wellcome Library, London, CC BY 4.0.

This is a good shock for our purposes. The American Civil War was not caused by anything happening in Lancashire machine shops, and it changed the relative supply of two inputs sharply and at a known date.

The shock

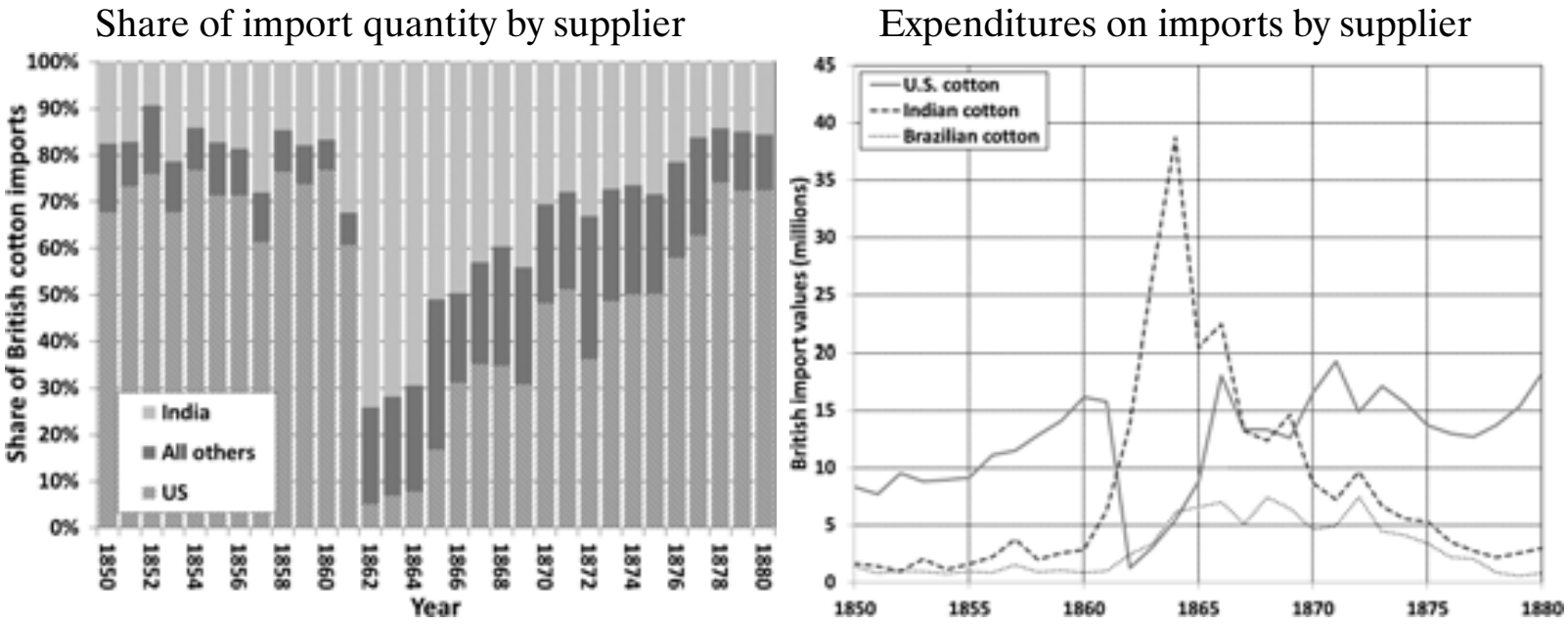


FIGURE 3.—British cotton import shares and expenditures, by supplier, 1850–1880. Data from

Hanlon (2015), Figure 3. British cotton import shares and expenditures by supplier, 1850–1880. The US share collapses from 1861 and the Indian share takes its place. American imports stay low through 1870.

Where Surat cotton broke the production line

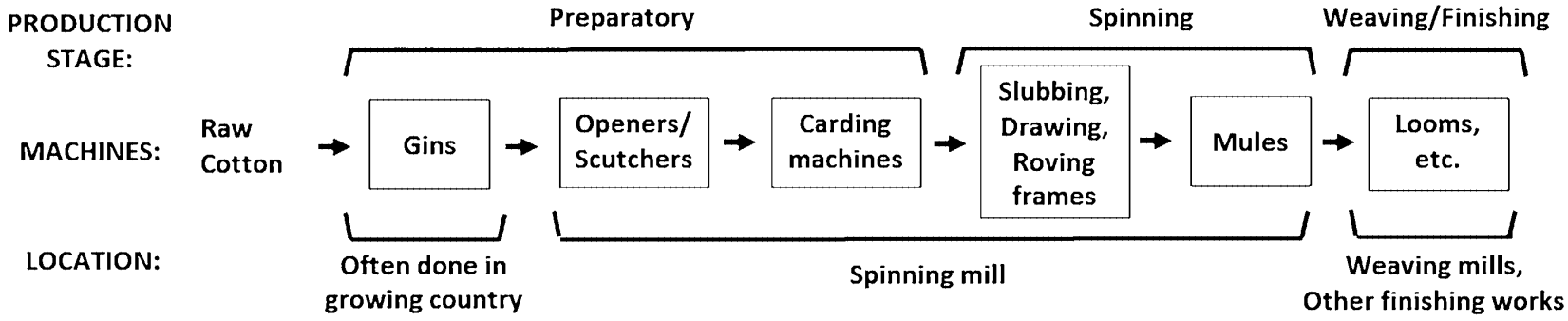


FIGURE 1.—The cotton textile production process.

Hanlon (2015), Figure 1. Raw cotton passes through gins, then openers and scutchers, then carding machines, before anything is spun. The treated classes are gins, openers and scutchers for cotton, and carding for dirt removal. The other nineteen subcategories of preparatory and spinning machinery, including the frames and mules to their right, are the controls.

Why Surat cotton was a problem

Indian cotton had a shorter, dirtier, more brittle fiber, and it broke the front end of the production line.

- **Gins** separate fiber from seed. **Openers and scutchers** open and clean the fiber. **Carding** machines comb it and remove dirt. Each of these choked on Surat.

- That specificity is what makes the prediction sharp. If invention responds to input supplies, patenting should surge in those three machine classes and not in the other nineteen subcategories of preparatory and spinning technology.

This is close to a pre-registered prediction. Contemporaries named the three bottleneck stages before anyone counted patents. The finer splits, cotton against other fibers and dirt removal against other carding, came from Hanlon reading the patent abstracts, and the other nineteen categories are the control group.

The response

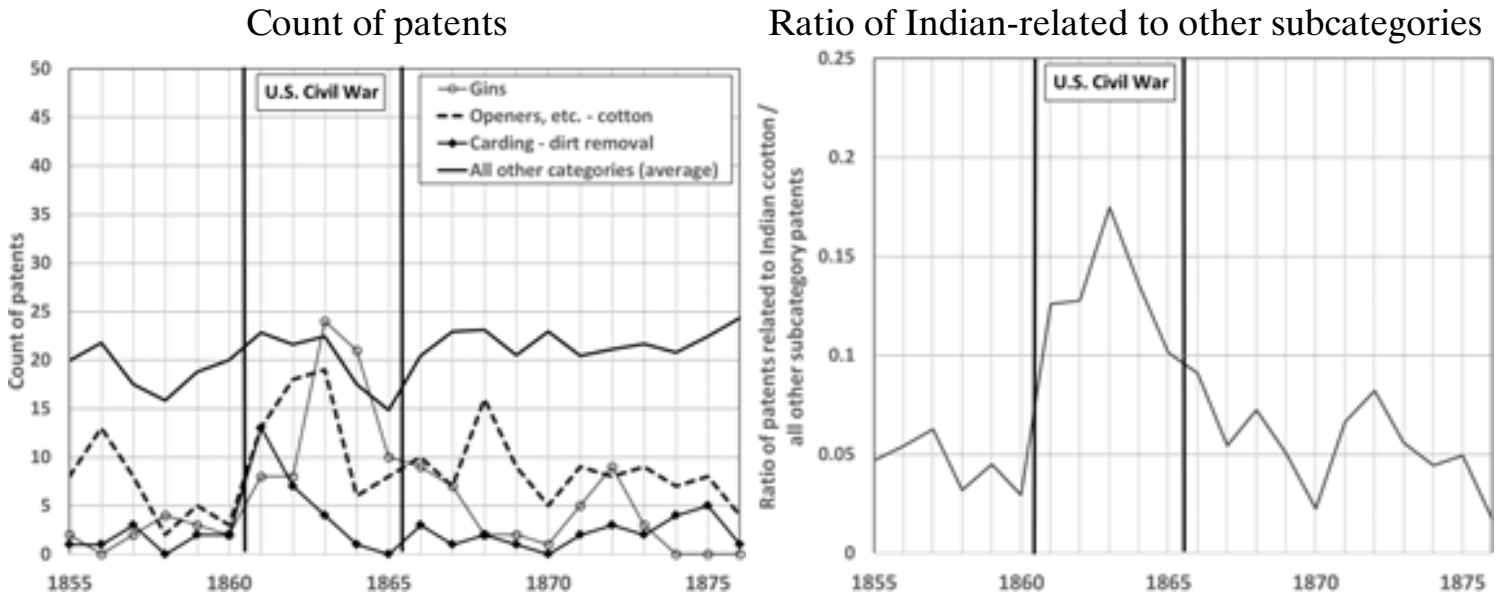


FIGURE 4.—Comparing Indian-related patents to other Preparatory & Spinning patents.

Hanlon (2015), Figure 4. Patents in the three Indian-cotton-related machine classes against the average of all other preparatory and spinning subcategories. The war years are boxed. Left panel counts, right panel the ratio of Indian-related to all other subcategory patents.

The specification

$$\log(\text{PAT}_{jt}) \;=\; \alpha \;+\; \beta\,(S_t \times \text{INDIATECH}_j) \;+\; \Psi_j \;+\; \xi_t \;+\; e_{jt}$$

- PAT_{jt} is the average patent count in technology subcategory j in period t. The periods are **1855–1860**, the shock years **1861–1865**, and **1866–1876**.
- INDIATECH_j marks the three subcategories that Surat cotton broke — gins, openers and scutchers, carding. The other nineteen are the controls.
- Ψ_j is a subcategory fixed effect, ξ_t a period effect.

Strip the notation away and this is a **difference in differences**. Two groups, before and after, and β on the interaction is the estimate. It is the design we worked at the board in Week 2, applied to patent counts instead of schooling.

Note what the fixed effects buy. Ψ_j absorbs the fact that some machine classes are simply patented more than others. ξ_t absorbs anything that moved all patenting at once, such as the war's effect on the whole industry.

The result, and three ways of judging it

TABLE I

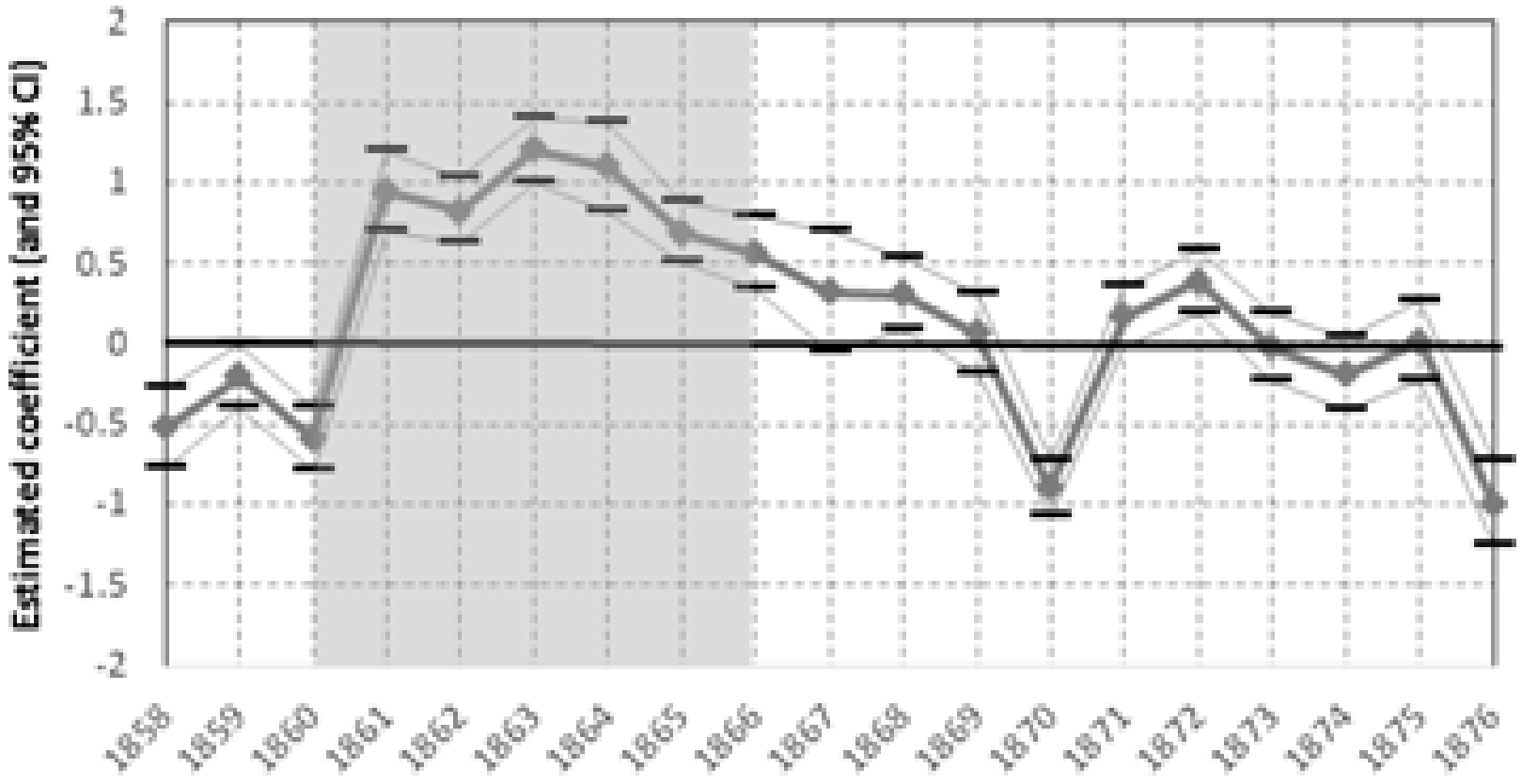
RESPONSE OF TECHNOLOGIES RELATED TO INDIAN COTTON DURING THE CIVIL WAR[a]

	Dependent Variable: Log Patents	
	Comparing Shock Period to Pre- and Post-Periods	Comparing Shock to Pre-Period Only
India-related	1.127	1.254
× Shock period	(0.276)	(0.302)
	[0.001]	[0.001]
	[[0.044]]	[[0.049]]
Subcategory effects	Yes	Yes
Time period effects	Yes	Yes
Observations	66	44
Number of panels	22	22

- The estimate is **1.127** in logs, so patenting in the affected classes roughly **tripled** relative to the controls during the war.
- Because the outcome is in logs, the multiple is $e^{1.127} \approx 3.1$. During 1861–1865 patents in the three India-related classes ran about 3.1 times their level before and after the war, over and above the change in the other 19 classes.
- Three numbers sit under it. **(0.276)** is a robust standard error. **[0.001]** is a **permutation** p-value. **[[0.044]]** is a small-sample correction using HC2 errors against a t distribution.

Hanlon (2015), Table I. Only 22 panels, so he does not trust asymptotics and says so.

The same result year by year



Hanlon (2015), Figure 5. The coefficient on Indian-related technology interacted with each year, 1858–1876, with 95 percent bands from Newey–West standard errors. The war years are shaded. Below zero before the war, which Hanlon ties to the disruption of Indian supplies after the 1857 Rebellion, then a jump in 1861, a peak in 1863, and a slide back to zero by 1869.

Why he ran a permutation test

- There are only **22** technology subcategories. With that few panels, standard errors built on asymptotics are likely to be too small, and a confident-looking result may not be one.

- So he takes every possible choice of three subcategories out of the twenty-two — $\binom{22}{3} = 1{,}540$ of them — pretends each is the treated group, and re-estimates.

- Under the null of no effect, those 1,540 placebo coefficients have the same distribution as the real one. Where the real estimate falls in that distribution is an exact p-value.

This is the **placebo logic** of Conley and Kelly from Week 3, run by the author himself, before a referee could make him. Assign the treatment at random many times, and see how often you get what you got.

Where the correction bites

TABLE II

HIGH-QUALITY PATENTS RELATED TO INDIAN COTTON DURING THE CIVIL WAR[a]

	Dependent Variable: Log Patents		
	Patents Paying Renewal Fee After Three Years		Patents Mentioned in *Newton's London Journal*
	(1)	(2)	(3)
India-related	1.343	1.805	0.887
× Shock period	(0.482)	(0.554)	(0.475)
	[0.002]	[0.001]	[0.052]
	[[0.125]]	[[0.086]]	[[0.204]]
Subcategory effects	Yes	Yes	Yes
Time period effects	Yes	Yes	Yes
Included time periods	Pre-shock, shock, post-shock	Pre-shock, shock	Pre-shock, shock

- These restrict to high-quality patents — those whose renewal fee was paid after three years, and those the trade press thought worth writing about. Two of the three point estimates are larger, up to 1.805.

- But look at the double brackets. Under the conservative small-sample correction the p-values are 0.125, 0.086 and 0.204. These robustness results do not clear five percent.

That is an honest table, and the honesty is the lesson. The main result survives every correction. The supporting results do not, and Hanlon prints them anyway.

What the numbers say

- Before the war, Indian-related machinery was about **4 to 6 percent** of patents in these subcategories.
- During 1861–1865 the share rose above **15 percent**, peaking in 1863.
- By 1867 it was back to **6 percent**. The redirection was sharp, specific, and it reversed when the reason for it went away.
- The internal timing fits the machine shops. Carding patents move first, openers and scutchers peak in 1862–63, gins not until 1863–64. Carding and opener patents modified machines already standing in the mills. A gin was a new machine that British makers had never built, since American cotton arrived already ginned.

For scale, textile innovations were over 11 percent of all British patents between 1855 and 1883. Textile innovation sat near the center of British patenting.

The part that makes it a theory test

- Directing invention at Indian cotton is interesting. What makes the paper an *Econometrica* paper is what happened to the price.

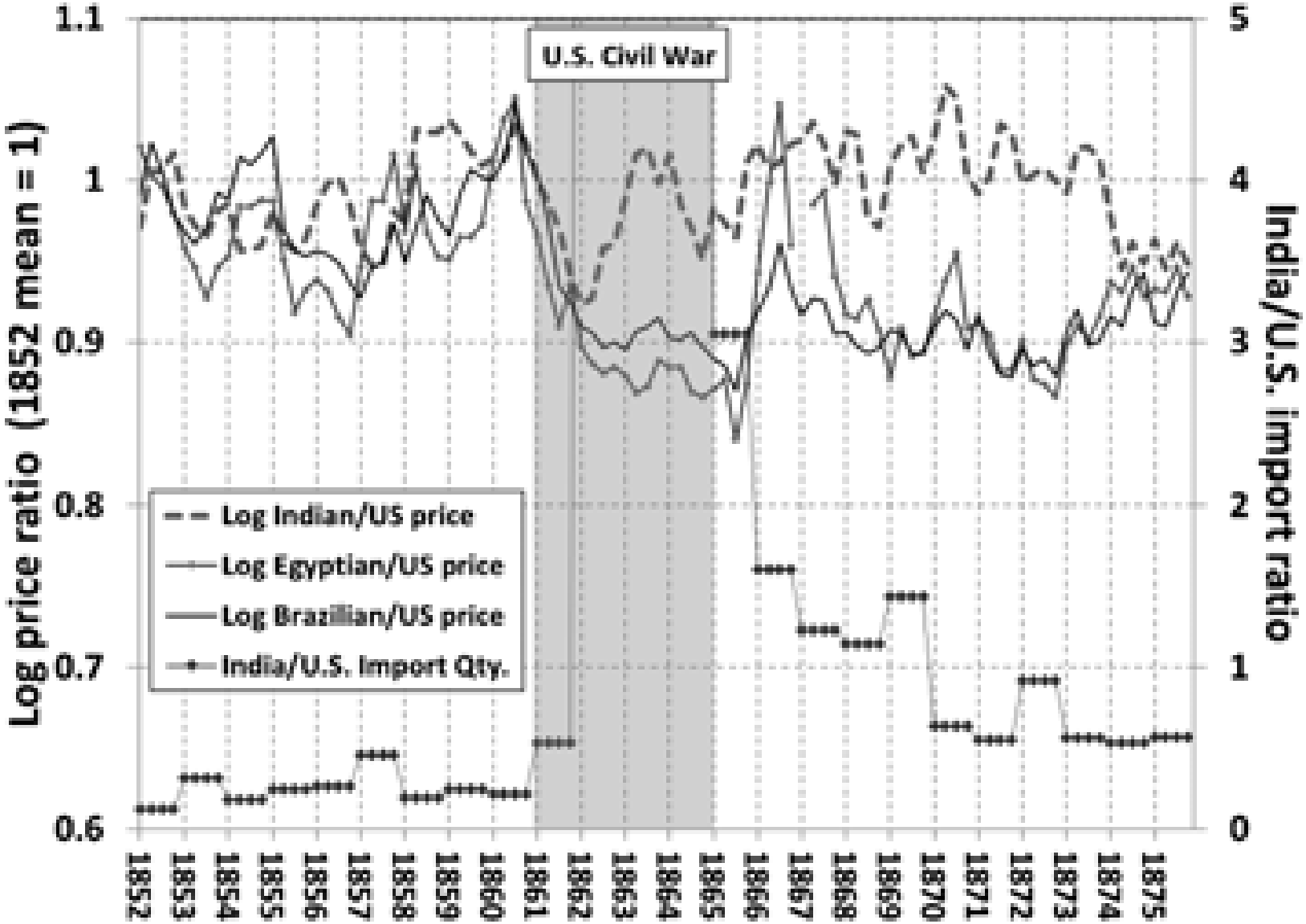
- As the new machines arrived, the relative price of Indian to American cotton rebounded to its pre-war level — despite Indian cotton being far more abundant relative to American than before.

- Ordinarily more supply means a lower relative price. Here technology made the abundant input better, and the price recovered.

The strong induced-bias hypothesis

When the relative supply of an input rises, induced technical change can raise its relative price back — a positive relationship between relative quantity and relative price. Directed technical change theory predicts it. This is the first well-identified test, and Hanlon says the pattern is consistent with it, with the caveat that it rests on a handful of price series.

The relative price came back while the quantity stayed high



Hanlon (2015), Figure 9. Dashed line, the log price of Indian relative to American cotton, 1852 mean set to one. Stepped line on the right axis, the ratio of Indian to American imports. The price is back near its pre-war level by 1863 while the quantity ratio stays at least double its pre-war level through 1870.

What this buys, and what it does not

Allen's mechanism is real and it is fast. A change in relative input supplies redirected inventive effort within two years, visibly, in patents, in machines, and then in prices.

- So the answer to the evening's opening question is that inventive activity has a large demand side, and it responds to prices on a timescale of years rather than generations.

- But hold the Week 3 discipline. This is one industry, one episode, one country, over five years. External validity is an argument to be made.

- And note what it does not say. Nothing here promises that the machines, once invented, make everyone equally productive.

Why did it not spread?

In 1910 Britain still dominated the industry it invented

- A century after the first mills, Britain still had **40 percent** of the world's factory cotton spindles and **81 percent** of net exports of cotton yarn and cloth.

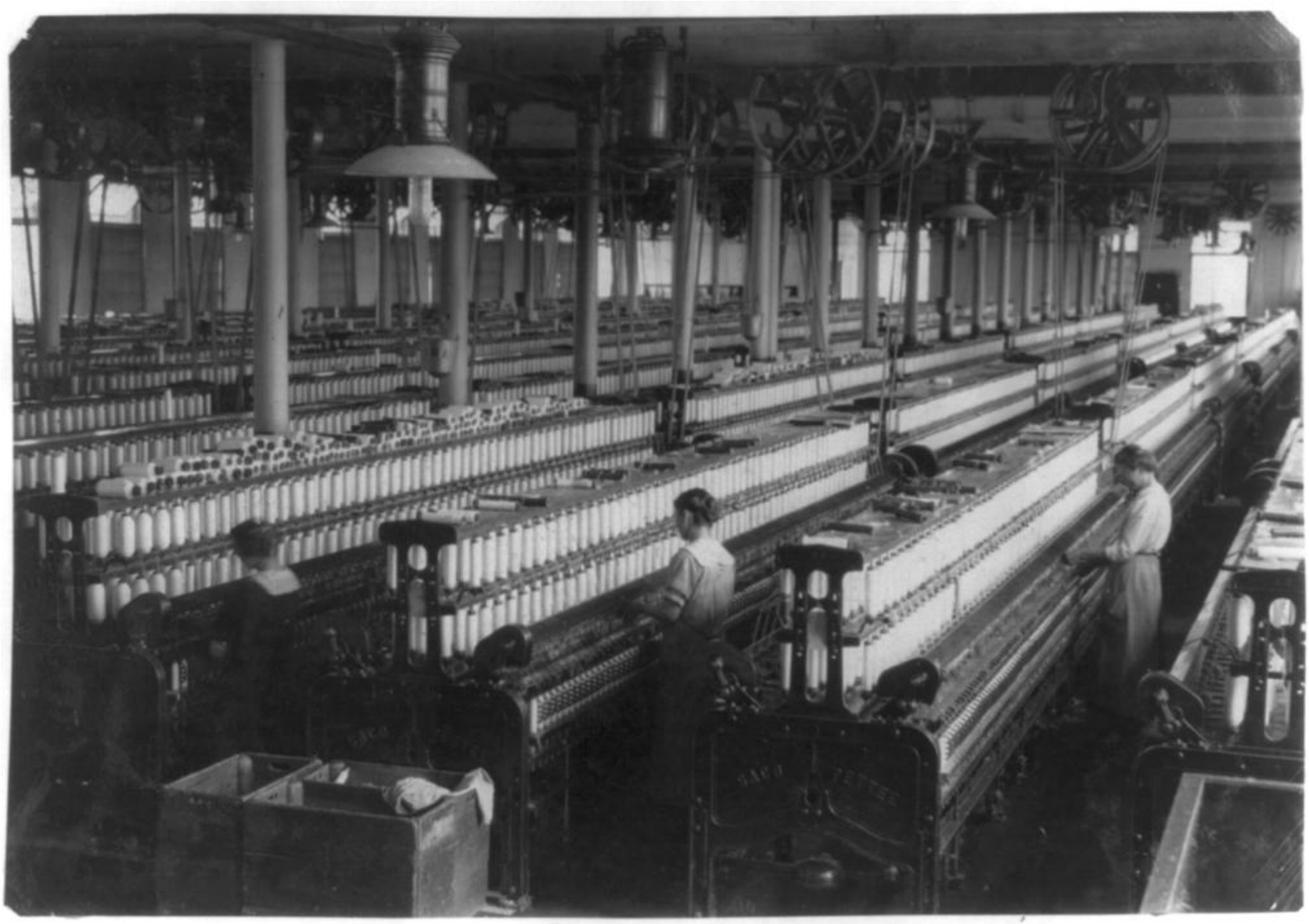
- And yet the technology was completely portable. The same machines, built in the same English factories, were installed everywhere from New England to Bombay, often under English managers, often newer abroad than at home because the industry was growing faster there.

- Producers elsewhere needed tariffs to survive against British goods, in their own markets, using British machines.

Same capital, same technology, same management, wildly different results. This is the residual that the entire subsequent literature on development is chasing.

Clark (1987), "Why Isn't the Whole World Developed? Lessons from the Cotton Mills."

What the puzzle looks like



A spinning room at the Indian Orchard mill, Massachusetts, 1916. Count the operatives, then count the frames. The same machines stood in Bombay and in Athens, tended by nearly six times as many people. Library of Congress, public domain.

What the input prices predicted

TABLE 1
COTTON TEXTILE COSTS AND PROFITS AS IMPLIED BY INPUT PRICES, c.1910

Country or Region	Weekly Wage Rate	Plant and Machinery (dollar/spindle)	Coal (dollar/ton)	Manufacturing Cost (England = 1.00)	Implied Profit Rate
New England	$8.8	$17.43	$3.80	1.59	−8.9%
United States (South)	6.5	17.43	3.80	1.30	−0.7
England	5.0	12.72	2.50	1.00	8.0
Germany	3.8	18.48	4.88	1.00	7.9
France	3.7	16.54	4.67	0.95	9.5
Switzerland	3.7	24.80	6.62	1.15	4.7
Austro-Hungary	2.8	16.38	5.75	0.85	12.6
Spain	2.7	19.33	6.50	0.91	10.5
Mexico	2.6	19.27	10.00	0.94	9.6
Russia	2.4	20.69	7.20	0.91	10.3
Italy	2.4	16.00	7.25	0.81	13.8
Portugal	1.72	17.50	7.00	0.76	15.0
Japan	0.80	24.57	2.58	0.73	14.1
India	0.78	17.56	5.02	0.61	19.1
China	0.54	16.32	3.25	0.53	22.1
Share in costs in England	0.618	0.124	0.034		

Note: The wage rate quoted is the average wage per worker per 55–hour week, not controlling for the sex and age composition of the labor force. The manufacturing cost in column 4 includes as the cost of capital a rate of return of 8 percent on the capital employed. The capital employed per spindle in each country is column 1 of Table 2. The return on capital constituted the remaining 22.4 percent of manufacturing cost in England.
Sources: See Appendix.

Clark (1987), Table 1. Input prices around 1910 and the manufacturing cost and profit rate they imply if workers were equally efficient everywhere. On these numbers India and China should have undercut England by 40 to 50 percent at profit rates above 19 percent, and New England should have lost money.

Machines per operative, about 1910

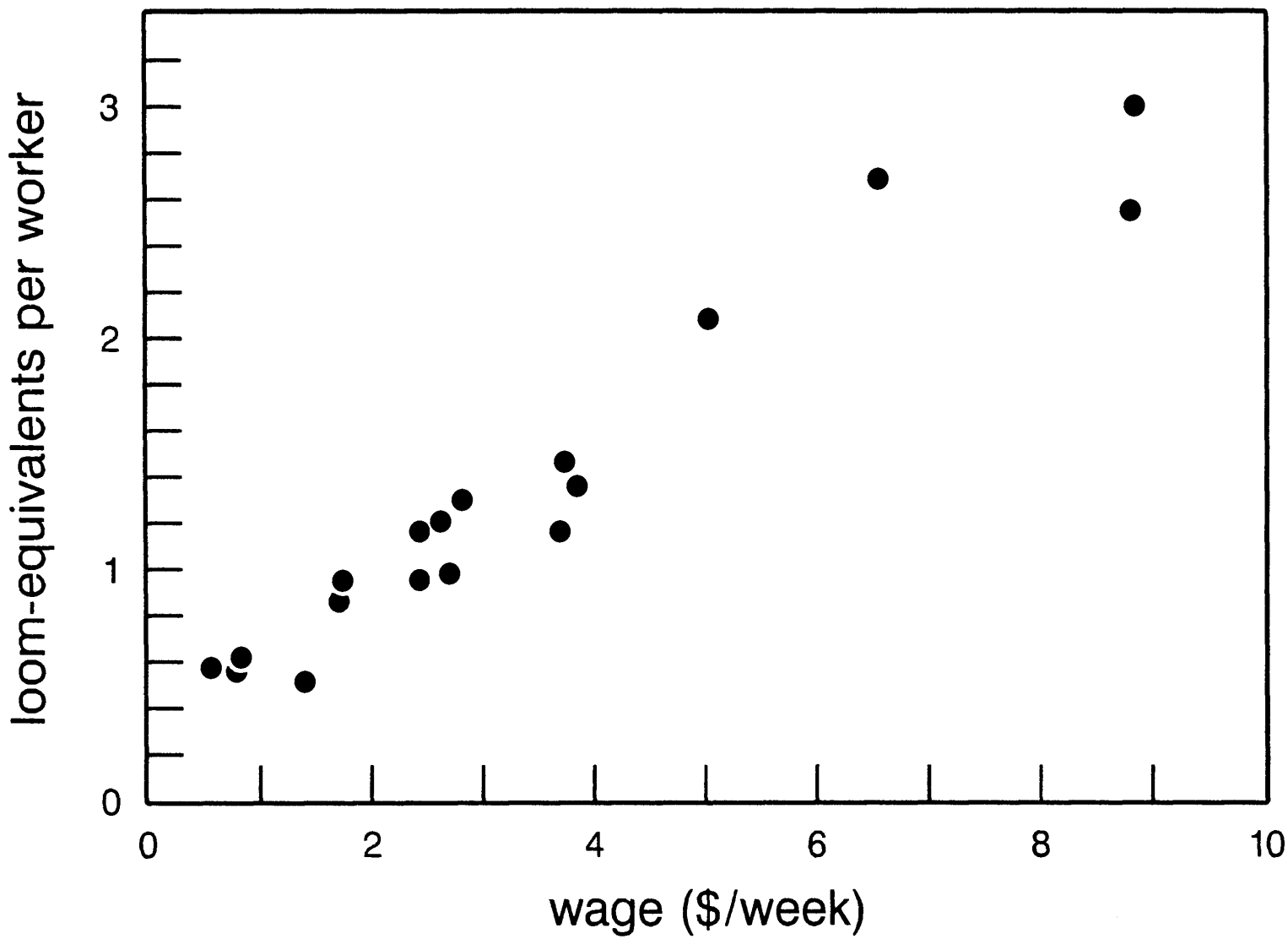
MACHINES PER OPERATIVE, c.1910

Country or Region	Average Weekly Wage	Loom-Equivalents per Worker	Index of Machines per Worker	Ring Spindles per Worker	Plain Looms per Worker
New England	$8.8	2.97	1.55	902	8.0
Canada	8.8	2.53	1.41	750	6.0
United States (South)	6.5	2.65	1.44	770	6.0
Britain	5.0	2.04	1.00	625	3.8
Germany	3.8	1.28	0.63	327	2.9
France	3.7	1.11	0.81	500	2.8
Switzerland	3.7	1.40	0.70	450	2.7
Austro-Hungary	2.8	1.24	0.65	403	2.8
Spain	2.7	0.91	0.73	450	2.0
Mexico	2.6	1.15	0.77	540	2.5
Russia	2.4	1.10	0.77	450	2.0
Italy	2.4	0.88	0.76	436	2.0
Portugal	1.72	0.88	0.65	384	2.0
Egypt	1.69	0.81	0.39	240	1.5
Greece	1.38	0.46			
Japan	0.80	0.53	0.52	190	1.6
India	0.78	0.50	0.33	214	1.9
China	0.54	0.48	0.34	168	1.5
Peru		1.17	0.78	391	3.5
Brazil		0.88	0.67	527	3.0

Notes: The United States and Canada used underpick looms and these were somewhat slower than the standard loom used elsewhere. In Brazil and Peru the nominal wages clearly exceeded the real wage greatly, but no price deflator is available

Clark (1987), Table 4. Read the index column. New England 1.55, Britain 1.00, Germany 0.63, India 0.33, China 0.34. And read the wage column beside it — the ranking is nearly the same.

Efficiency tracks the local wage



Clark (1987), Figure 1. Each point is a country. Better-paid workers tended more machinery, and the relationship is tight. Which way the causation runs is the whole question Clark spends much of the paper on.

The gap was more than six to one at the extreme

> *"In 1910 one New England cotton textile operative performed as much work as 1.5 British, 2.3 German, and nearly 6 Greek, Japanese, Indian, or Chinese workers."*

- Differences in worker efficiency of about **6.5 to 1** at the extreme.

- Clark's method is to enumerate the explanations an economist would reach for and kill them one at a time.

Suspect	**Verdict**
Different technology	same machines, English-built, often newer abroad
Price of raw cotton	transport mattered only in Mexico and Switzerland
Non-labor conversion costs	outweighed by the labor cost advantage
Capital utilization	would have helped the others rather than Britain
Capital-labor substitution	checked directly, does not account for it
Cotton quality substitution	overstaffing appears where quality is irrelevant
Worker experience	experience gaps too small, gradient too flat
Inherent worker quality	killed by the migration evidence

The cost consequence

IMPLIED COSTS AND PROFITS IN COTTON TEXTILES ADJUSTING FOR WORKER EFFICIENCIES, c.1910

Country or Region	Weekly Wage Rate	Machinery per Worker (loom-equivalent)	Corrected Yearly Labor Cost	Manufacturing Cost (England = 1.00)	Implied Profit Rate
New England	$8.8	2.97	$6.04	1.25	0.9%
Canada	8.8	2.53	7.10		
United States (South)	6.5	2.65	5.00	1.12	4.6
England	5.0	2.04	5.00	1.00	8.0
Germany	3.8	1.28	6.06	1.28	0.1
France	3.7	1.11	6.80	1.33	−1.9
Switzerland	3.7	1.40	5.39	1.36	0.0
Austro-Hungary	2.8	1.24	4.61	1.07	5.8
Spain	2.7	0.91	6.05	1.32	−0.7
Mexico	2.6	1.15	4.61	1.19	2.9
Russia	2.4	1.10	4.45	1.16	3.8
Italy	2.4	0.88	5.56	1.20	1.8
Portugal	1.72	0.88	3.99	1.04	7.0
Egypt	1.69	0.81	4.26		
Greece	1.38	0.46	6.12		
Japan	0.80	0.53	3.08	1.01	7.7
India	0.78	0.50	3.18	0.91	10.6
China	0.54	0.48	2.30	0.75	15.5

Note: The implied costs and profit rates in this table are based on input prices alone and do not allow for differences in capital utilization. Brazil and Peru are omitted from this table, though there

Clark (1987), Table 3. Look at the corrected labor cost column. England 5.00. Greece **6.12**, on a wage barely over a quarter of England's. Cheap labor was not cheap. Only India and China come out cheaper on these prices, Japan joins them once longer hours are counted, and those three were the only countries that competed with Britain.

How Clark actually argues it

Clark has no regression either. His method is an **accounting decomposition**, laid out here in four steps.

Step	What it does
1. Write down the cost identity	raw cotton is about two thirds of total cost. Of the rest, wages are about two thirds
2. Price every input, everywhere	15 countries, around 1910, from consular and tariff-board reports
3. Kill each candidate in turn	technology, cotton price, capital, utilization, substitution, experience, worker quality
4. Name what is left	the residual is the finding

A regression puts the residual in the error term and reports the coefficients. Clark does the reverse. He measures every coefficient he can and then points at the residual, because the residual is what he is arguing about.

The migration evidence

If the difference were in the workers themselves, moving them should not help. It did.

- In 1911, **27.8 percent** of workers in northern American cotton mills were Polish, Portuguese, Greek or Italian — from countries needing three to six times as many workers per machine. In New England they were efficient.

- Brazilian mills hired Portuguese, Italians and Germans and got operatives about **one third** as efficient as New England's.

- Earnings of immigrant groups in American manufacturing barely differ by country of origin once corrected for age.

"Why did the Irish who happened to get on the boat to Lancashire exhibit lower efficiency as cotton operatives than those who instead took the boat to New England?"

Same people, two boats. Whatever it is, it belongs to the place.

Immigrant earnings barely varied by origin once age is held fixed

TABLE 8
EARNINGS OF U.S. MALE IMMIGRANTS IN MANUFACTURING, 1910

Country of Origin	Efficiency of Textile Workers (home country)	Average U.S. Wage (England = 100)	Average Age	Age–Adjusted Wage (England = 100)
Canada	124	78	39.0	75
England	100	100	38.7	100
Germany	63	96	40.9	90
France	54	93	36.1	98
Switzerland	69	99		
Austro-Hungary	61	85	31.6	93
Spain	45	106	31.4	115
Mexico	56	90	32.7	102
Russia	54	80	30.7	95
Italy	43	80	29.6	98
Portugal	43	57	26.4	79
Greece	23	60	26.5	83
Japan	26	75	29.5	93

Note: The adjusted earnings are derived by regressing earnings on the average age, and then subtracting the estimated effect of age on earnings. Estimated in this way, age explains 44 percent of the variance of the reported earnings.

Sources: The earnings and average age of immigrants are from U.S. Senate, Immigration Commission, *Abstracts of the Reports of the Immigration Commission* (Washington, D.C., 1911), table 22, p. 367; table 26, p. 371; table 88, pp. 463–64.

Clark (1987), Table 8. Efficiency of textile workers at home against the age-adjusted earnings of male immigrants from the same countries in American manufacturing, 1910, England equal to 100 in both. Greece 23 at home and 83 in America, Italy 43 and 98, Portugal 43 and 79.

What it looked like on the mill floor

"The operatives in this mill refuse to attend to more machinery. I watched two ring frames for three minutes; there was not a single end down, yet the workpeople would not look after more than one side. They said that they are satisfied with the present wage, and that there are so many men who want work and cannot get it that it would be unfair if they were to attend to more machines."

"...it was apparent that they could easily have taken care of more, but they won't... They cannot be persuaded by any exhortation, ambition, or the opportunity to increase their earnings."

- Bombay, as recorded by Pearse, and an American visitor in the 1920s. When management tried to raise machines per worker in 1928, the result was the great Bombay mill strike.

Read the workers' stated reason again. It is an argument about fairness under unemployment.

What Clark concludes, and what he does not

- His conclusion is that local culture seems to have determined worker performance — norms about effort, manning and fairness that bound the whole local labor force.

- He is not saying anyone was lazy, and he writes that the sources "do not allow me to go with any confidence beyond this limited ascription of responsibility to local influences." Whatever constrained the choices applied to everybody in that labor market, and the same people behaved differently in a different one.

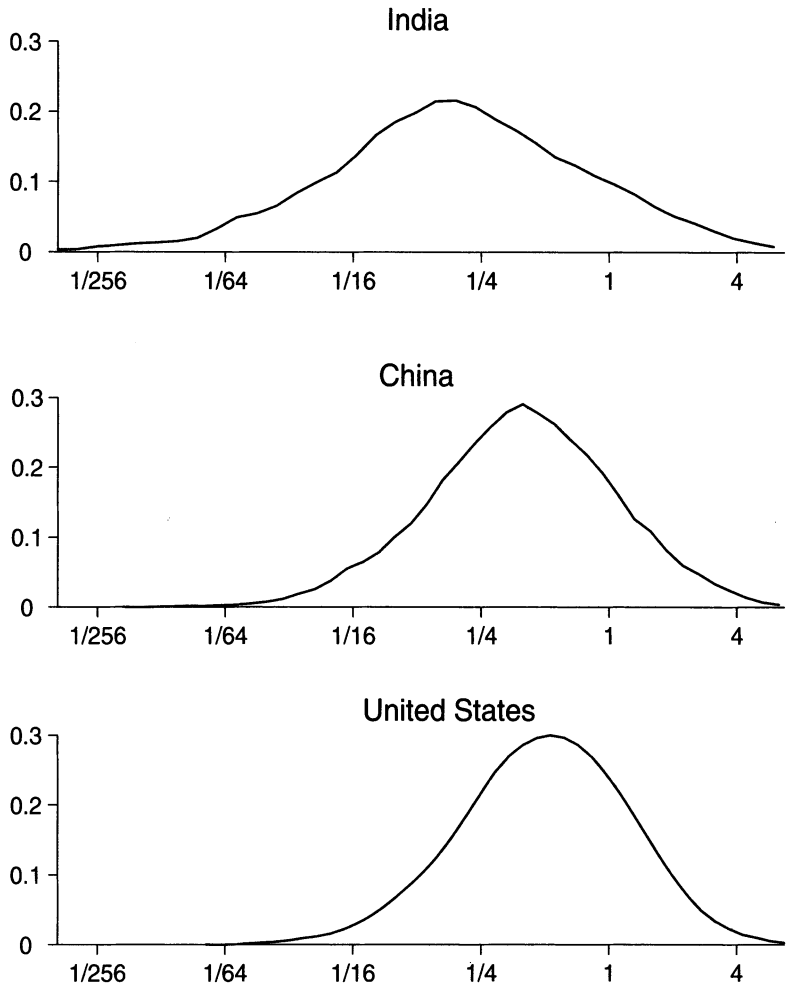
Technology transfers at the speed of a mail ship. The social arrangement that makes technology productive transfers at the speed of culture — which is Week 4's lesson, standing on a factory floor.

Three consequences Clark draws from the mills

- **The wage misprices labor.** A mill buys machine-tending, so the true price of labor is the wage divided by the work each operative does. Greek mills paid 28 percent of England's wage for 23 percent of the work, which left their labor cost per unit of output at 6.12 against England's 5.00. Across countries wages varied 16 to 1 and true labor cost 3 to 1.
- **Workers move to the capital.** Productivity belongs to the place. A mill moved to Athens inherits Athenian manning, and a worker moved to Massachusetts takes on Massachusetts manning. Greek operatives ran at 23 percent of English efficiency at home, and Greek immigrants in American manufacturing earned 83 percent of what English immigrants earned. Nearly 10 million Europeans left for the United States in 1903–1910, most of them for manufacturing.
- **Efficiency rises when norms change.** English weavers tended 2.2 looms in 1850 and 3.44 in 1906, and the 1906 looms ran about 50 percent faster. Clark argues that the force separating countries in 1910 could also operate over time, so part of that growth may be work intensification.

Migrants chose to leave, so part of the move from 23 to 83 could be selection. Clark answers that poorer countries sent proportionally more emigrants, and that Irish operatives who took the boat to Lancashire ran below those who took it to New England.

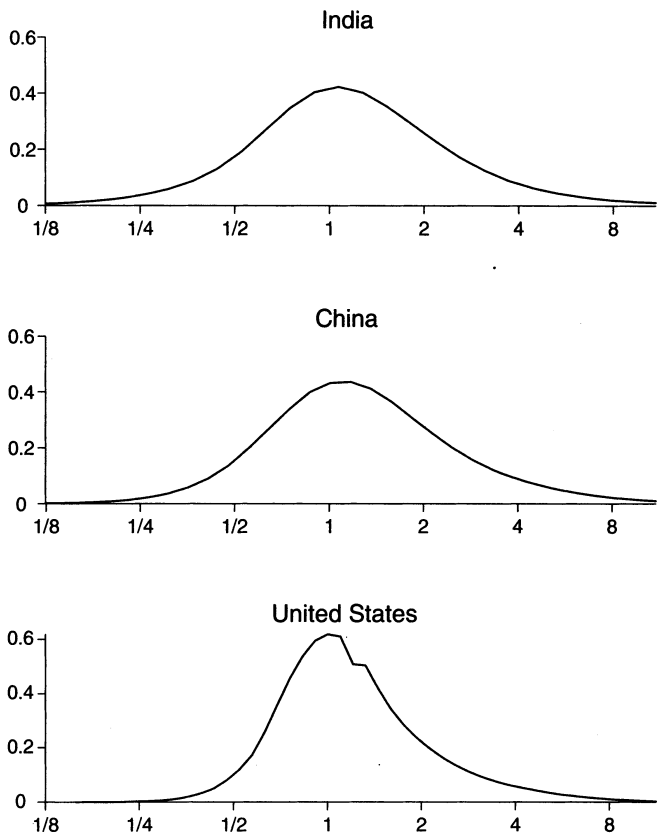
Physical productivity across plants in the same industry



- Hsieh and Klenow (2009) use manufacturing censuses for India in 1994, China in 2005 and the United States in 1997. Each plant is measured against the mean of its own narrowly defined industry.
- **TFPQ** is physical productivity, output per unit of capital and labor. Clark measured one industry's national average. Here the whole distribution within industries is on view.
- India's left tail is far thicker. The ratio of the 75th to the 25th percentile is 5.0 in India, 3.6 in China and 3.2 in the United States. Hsieh and Klenow read the tail as policies that keep inefficient plants alive.

Hsieh and Klenow (2009), Figure I. Log scale. China's survey undersamples small private plants.

Marginal products are unequal across plants in China and India



- **TFPR** is price times physical productivity. Without distortions a more productive plant grows and cuts its price until its TFPR matches its rivals'. Its spread measures the wedges on capital and labor.
- The ratio of the 90th to the 10th percentile of TFPR is 5.0 in India, 4.9 in China and 3.3 in the United States.
- Equalizing TFPR to the American degree would raise manufacturing TFP 30–50 percent in China and 40–60 percent in India. That is roughly half the TFP gap with the United States for China and a third for India.

Both figures measure each plant against its industry's mean. A norm that binds every mill in a place, as Clark's did, shifts the whole distribution and is invisible here. It is a candidate for the part of the gap that misallocation leaves unexplained.

Hsieh and Klenow (2009), Figure II. Measurement error also spreads TFPR, and Bils, Klenow and Ruane (2021) find it inflates India's dispersion.

Where this leaves us

The three papers side by side

	Question	**Answer**
Allen	why invented in Britain?	relative prices made it pay there
Hanlon	does invention chase prices?	yes, causally, within two years
Clark	does diffusion equalize?	no, the labor side does not travel

- Two tensions worth naming rather than smoothing over.

- Allen says high wages were Britain's advantage, because they induced labor-saving invention. Clark says high wages went with high efficiency, so the wage was never the burden it looked like. These are the same fact seen from two sides, and reconciling them is a good oral exam question.

- Hanlon proves that invention responds to input scarcity — in the very industry where Clark shows the binding constraint abroad was not an input you could buy.

Three questions and three different methods

	Method	**Where the evidence lives**
Allen	prices plus a model	cross-sectional price series
Clark	accounting decomposition	the residual, after killing suspects
Hanlon	difference in differences	β on an interaction, with fixed effects

- Only one of tonight's three papers has a coefficient and a standard error. Allen wrote in 2011, four years before Hanlon, and had no use for one.

- That is a story about what each question will bear. You cannot run a difference in differences on the eighteenth century, because nobody recorded a control group.

- But notice the cost. Hanlon can tell you the effect of one shock on one industry over five years, precisely. Allen is trying to explain the industrial revolution. The sharper the identification, the narrower the question it answers.

Hold that trade-off. It is the central tension of the whole field, and it will come up in the oral exams.

How to read an industrialization paper

The weekly checklist. As with Weeks 3 and 4, these get asked in the orals.

1. **Which margin?** Invention, adoption, or productivity in use. Papers that blur these are usually wrong somewhere.
2. **Which relative price** does the argument turn on, and is it measured or asserted?
3. **Would it survive a shock test?** Could you find a Hanlon-style experiment for this mechanism, and what would it show?
4. **What is the residual made of?** If output differs with identical inputs, name what is left, and say how you would measure it.
5. **Whose behavior is doing the work?** Inventors, firms, or workers.

Question four is where Clark starts and where most of development economics still lives.

Next week

- **Week 6, 5 October. Cities.**

- Davis and Weinstein on bones, bombs and break points — what happens to a city's population when it is destroyed, and what that reveals about why cities sit where they do.
- Bleakley and Lin on portage sites, and economic activity that persists long after the reason for it is gone.

The factories had to stand somewhere. Next week asks why activity piles up where it does, whether the piles can be moved, and what happens when somebody tries.